

Daily Logic Drill #2

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Implication, Converse, Inverse & Contrapositive · Necessary vs Sufficient.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Rapid condition strength: classify as necessary, sufficient, both (equivalent), or neither.

- For real numbers x, y , classify “ $x > y$ ” as a condition for “ $x^3 > y^3$ ”.
- For non-zero real numbers x, y , classify “ $x > y$ ” as a condition for “ $\frac{1}{x} < \frac{1}{y}$ ”.
- For the quadratic equation $x^2 - 2kx + 1 = 0$, classify “ $k > 1$ ” as a condition for “the equation has two distinct positive real roots”.
- For real numbers x, y , classify “ $|x + y| = |x| + |y|$ ” as a condition for “ $xy \geq 0$ ”.
- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable. Classify “ $f'(x) > 0$ for all $x \in \mathbb{R}$ ” as a condition for “ f is strictly increasing on $\mathbb{R}$ ”.
- For real constants a, b, c, d , classify “ $ad - bc \neq 0$ ” as a condition for “the system $ax + by = 0$, $cx + dy = 0$ has only the trivial solution $(x, y) = (0, 0)$ ”.
- For a point $(x, y) \in \mathbb{R}^2$, classify “ $x^2 + y^2 \leq 1$ ” as a condition for “ $|x| + |y| \leq \sqrt{2}$ ”.
- For a positive integer n , classify “ $12 \mid n$ ” as a condition for “ $4 \mid n$ and $6 \mid n$ ”.
- For a triangle with side lengths a, b, c , classify “ $a^2 + b^2 = c^2$ ” as a condition for “the area of the triangle is $\frac{1}{2}ab$ ”.
- For an integer n , classify “ n is odd” as a condition for “ $n^2 \equiv 1 \pmod{8}$ ”.

Section B Manipulation Drills

(≤50 s each)

Condition strength on roots, calculus symmetries, and number theory.

- Determine the necessary and sufficient condition on real k such that the quadratic equation $x^2 - kx + 9 = 0$ has two distinct real roots that are both strictly greater than 1. (*after TMUA 2021 Paper 2 Q9*)

2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable. Consider the conditions:
 - I. $f(-x) = f(x)$ for all $x \in \mathbb{R}$.
 - II. $f'(-x) = -f'(x)$ for all $x \in \mathbb{R}$.
 Classify condition I as a condition for condition II. (*after TMUA 2018 Paper 2 Q10*)
3. For positive real numbers x, y , classify “ $\ln(x + y) < \ln x + \ln y$ ” as a condition for “ $x > 1$ and $y > 1$ ”.
4. State the contrapositive of: “If $a^2 + b^2$ is divisible by 3, then both a and b are divisible by 3” (for integers a, b). Is this contrapositive true?
5. Let T_1 and T_2 be two triangles. Classify “ T_1 and T_2 have equal perimeter and equal area” as a condition for “ T_1 and T_2 are congruent”.
6. For an integer n , classify “ n is odd” as a condition for “ $n^3 - n$ is a multiple of 24”.
7. Suppose propositions P, Q, R satisfy $P \implies Q$ and $R \implies Q$. Must either $P \implies R$ or $R \implies P$ be true? Justify with a mathematical counterexample.
8. For real c , determine whether “ $c < 0$ ” is necessary, sufficient, both, or neither for the quartic equation $x^4 + x^2 + c = 0$ to have at least two distinct real roots.
9. Consider the statement: “If n is a multiple of 9, then n^2 is a multiple of 27” (for positive integer n). State its converse, and determine whether the converse is true or false.
10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable. Classify “ $f'(a) = 0$ and $f''(a) > 0$ ” as a condition for “ f has a local minimum at $x = a$ ”.

Section C Substitution & Structure

(≤ 90 s each)

Chain necessary/sufficient claims, and use the inverse-converse link.

1. For real numbers x, y , the condition “ $x + y$ is an integer and $x - y$ is an integer” is:
 - A) necessary but not sufficient for “ x and y are both integers”
 - B) sufficient but not necessary for “ x and y are both integers”
 - C) necessary and sufficient for “ x and y are both integers”
 - D) neither necessary nor sufficient for “ x and y are both integers”
2. Determine the condition on real parameters p, q that is necessary and sufficient for the cubic $x^3 - 3px + q = 0$ with $p > 0$ to have three distinct real roots.
 - A) $p^3 > q^2$
 - B) $4p^3 > q^2$
 - C) $p^2 > 4q^3$
 - D) $q^2 > 4p^3$

3. Classify “a triangle has all three altitudes of equal length” as a condition for “the triangle is equilateral”. Justify using the fact that an altitude to side a has length $\frac{2 \times \text{Area}}{a}$.
4. For positive integers a, b, n , suppose you’re told only that n divides a , or n divides b (possibly both). Does this pin down whether n divides the product ab ? And working the other way: does knowing n divides ab pin down whether n divides a or n divides b ? (after TMUA 2019 Paper 2 Q5)
- A) “ n divides a or n divides b ” is necessary but not sufficient for “ n divides ab ”
- B) “ n divides a or n divides b ” is sufficient but not necessary for “ n divides ab ”
- C) “ n divides a or n divides b ” is necessary and sufficient for “ n divides ab ”
- D) “ n divides a or n divides b ” is neither necessary nor sufficient for “ n divides ab ”
5. For a twice-differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$, which of the following is necessary and sufficient for the midpoint convexity condition $\frac{f(a)+f(b)}{2} \geq f\left(\frac{a+b}{2}\right)$ to hold for all $a, b \in \mathbb{R}$? (after TMUA 2018 Paper 2 Q18)
- A) $f''(x) > 0$ for all $x \in \mathbb{R}$
- B) $f''(x) \geq 0$ for all $x \in \mathbb{R}$
- C) $f'(x) \geq 0$ for all $x \in \mathbb{R}$
- D) $f(x) \geq 0$ for all $x \in \mathbb{R}$
6. Classify “ n^2 is even” as a condition for “ n is even”.
- A) necessary but not sufficient
- B) sufficient but not necessary
- C) necessary and sufficient
- D) neither necessary nor sufficient
7. Let $p(x)$ be a cubic polynomial with real coefficients. Which of the following conditions is necessary and sufficient for the existence of $c \in (a, b)$ such that $p'(c) = 0$? (after TMUA 2021 Paper 2 Q8)
- A) $p(a) = p(b)$
- B) $p'(a)p'(b) \leq 0$
- C) $p'(x) = 0$ has real roots and at least one lies in (a, b)
- D) $p(a)p(b) < 0$
8. State the contrapositive of “If a and b are both irrational, then ab is irrational.” Is the original statement true? Give a counterexample if false.

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Let $f(x) = x^2 + bx + c$ with $f(0) = c$. Let P be the statement: “if $c < 0$, then f has two distinct real roots.” Which of the following must be true? (after TMUA 2022 Paper 2 Q5)

- I. P
- II. The converse of P
- III. The contrapositive of P
- A) none of them
- B) I only
- C) II only
- D) III only
- E) I and II only
- F) I and III only
- G) II and III only
- H) I, II and III
2. Consider a finite list of n real numbers (repeats allowed), sorted into non-decreasing order. Let P be the statement “ n is odd”, and let Q be the statement “the middle value of the sorted list equals one of the list’s own entries”. P is: (after TMUA 2022 Paper 2 Q6)
- A) necessary but not sufficient for Q
- B) sufficient but not necessary for Q
- C) necessary and sufficient for Q
- D) neither necessary nor sufficient for Q
3. For a positive integer n , let P be “ n leaves remainder 1 when divided by 4” and Q be “ n^2 leaves remainder 1 when divided by 8.” P is:
- A) necessary but not sufficient for Q
- B) sufficient but not necessary for Q
- C) necessary and sufficient for Q
- D) neither necessary nor sufficient for Q
4. Statement S : “If a quadrilateral has four equal sides, then its diagonals are perpendicular.” (a) Is S true? (b) State S ’s converse. Is the converse true?
5. Statement S : “If n is a positive integer such that $n^2 - 1$ is prime, then $n = 2$.” (a) Is S true? Prove or disprove. (b) State S ’s contrapositive. (c) Is “ $n^2 - 1$ is prime” necessary and sufficient for “ $n = 2$ ”?

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at [github.com/David Akinyele-Aje/speed-maths](https://github.com/David-Akinyele-Aje/speed-maths)